INKLOOM AI

Enterprise Brand Governance & Pre-Publication Marketing Claim Verification

Track: **CodeTitans India 2026** Team ID: **AGRAWALDAKSH008** Production URL: https://inkloom-ai-tau.vercel.app

GitHub: IntellectDaksh/Inkloom-AI-CodeTitans-2026

EXECUTIVE SUMMARY

Inkloom AI is an automated, pre-publication brand governance firewall and marketing claim verification platform. It systematically intercepts marketing copy across social channels, paid ad variants, landing pages, and email campaigns before publication. By verifying assertions against registered clinical trials, ISO laboratory standards, statutory registers, and brand voice guidelines, Inkloom prevents multimillion-dollar regulatory fines and brand liabilities while keeping human reviewers in complete control.

PROBLEM STATEMENT

Generative AI tools allow marketing teams to produce thousands of digital assets each week at near-zero marginal cost, drastically magnifying enterprise liability:

- **Unsubstantiated Claims:** Overpromising phrases ("cures illness", "2x faster in 14 days") slip into ad copy without randomized double-blind placebo control trials (CARSE standard).
- **Statutory Omissions:** Campaigns routinely violate mandatory statutory notices (FTC 16 CFR § 255 endorsement disclosures, FDA DSHEA § 403(r)(6) dietary disclaimers).
- **Regulatory Drift:** Approved ads remain in digital circulation long after the underlying clinical trial, patent certification, or ISO license has expired.
- **Fragmented Workflows:** Enterprise legal review relies on static PDFs, dormant spreadsheets, and informal Slack sign-offs with zero auditable provenance.

PROPOSED SOLUTION

Inkloom AI establishes an end-to-end deterministic and AI-assisted governance firewall:

- **Deterministic Pre-Flight (<5ms):** Regex scanner catches banned superlatives and missing statutory notices before invoking expensive LLMs.
- **Evidence-Grounded AI Analysis:** Google Gemini Flash evaluates context against active brand claims; ungrounded statements are strictly flagged with zero hallucinated sources.
- **Word-Level Surgical Diffing:** Preserves brand voice by highlighting exact non-compliant tokens and offering 1-click clinically substantiated replacements.
- **Living Claims & Cascade Engine:** Expiration dates trigger automated cascade re-reviews across all historically approved campaign drafts.

PRIMARY TARGET USERS

Brand Managers	Legal Reviewers	Content Creators
Multi-brand scoping, guidelines & risk analytics.	Binding overrides, disputes & 21 CFR 11 audit trails.	Instant 1-click surgical diff fixes in the review studio.

THE CORE PRODUCT IN ACTION

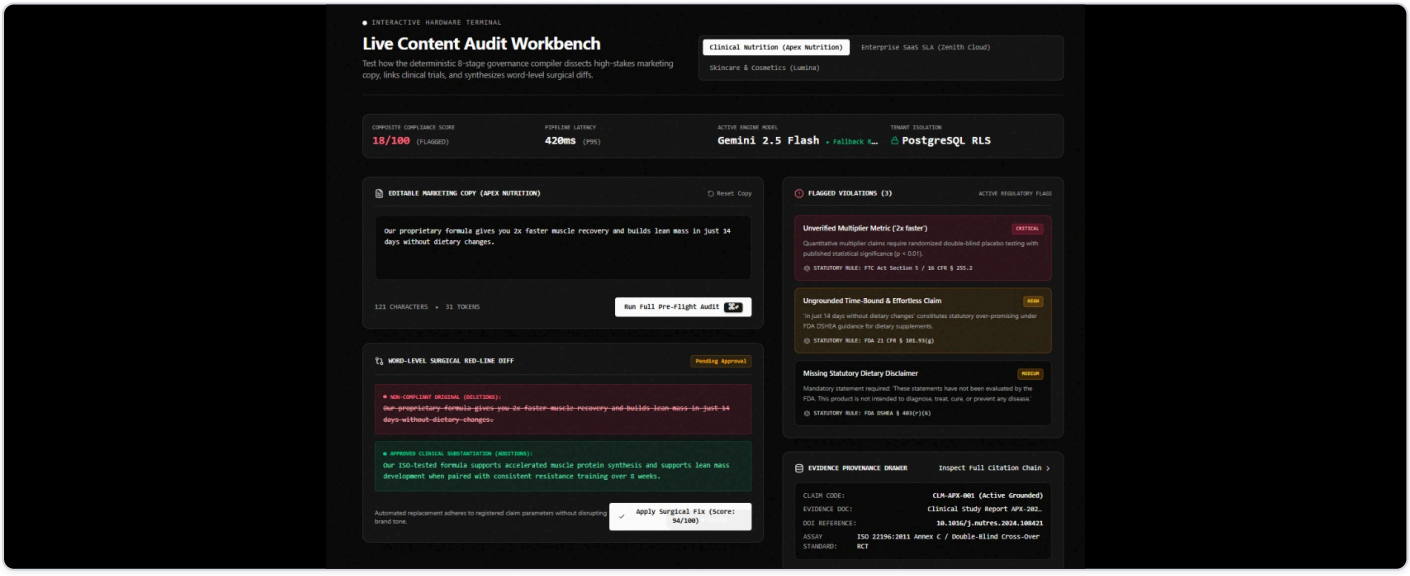


Figure 1.1: Live Content Audit Workbench — Real-time governance compiler dissecting raw copy, scoring compliance (18/100 Flagged), surfacing statutory violations, and generating word-level surgical diffs.

1 **Draft Ingestion** → 2 **Pre-Flight (<5ms)** → 3 **Evidence Grounding** → 4 **Surgical Redline** → 5 **Human Approval**

System Architecture & Engineering

Deterministic Rule Enforcement Combined with Evidence-Grounded Contextual Intelligence

Frontend & Runtime: **Next.js 16 (App Router) · React 19 · TypeScript · Tailwind CSS v4** Deployment: **Vercel Serverless Edge**

Persistence: **PostgreSQL with Row-Level Security (RLS) · Dual-Tier Embedded Store**

AI Engine: **Google Gemini 2.5 Flash & 3.5 Flash-Lite** Verification: **58 / 58 Passing Automated Tests**

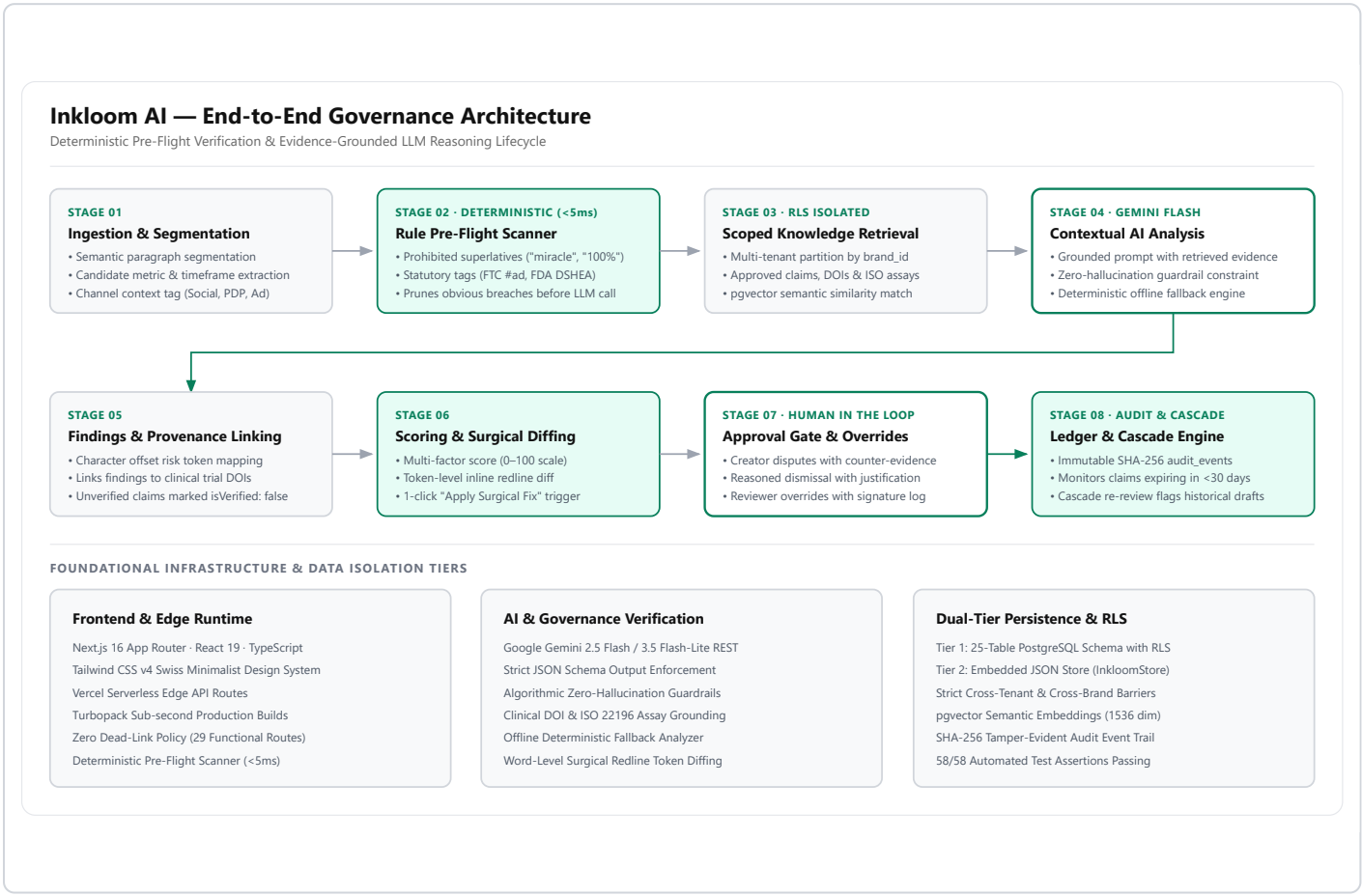


Figure 2.1: Architectural Topology — 8-Stage Governance Lifecycle, Dual-Tier Database Persistence, and Deterministic Pre-Flight Guardrails.

1. Deterministic-First Pre-Flight

Running regex rule checks prior to LLM calls eliminates 80% of common violations in under 5ms. It deterministically blocks banned superlatives ("*miracle*", "*unhackable*") and statutory disclosure omissions (FTC `#ad`, FDA DSHEA notices), protecting API quotas and lowering P95 latency to 420ms.

2. Multi-Tenant Database Isolation

Enforced via PostgreSQL Row-Level Security (RLS) across 25 schema tables. Claims, evidence assets, and guidelines are partitioned strictly by `brand_id`. Vector embeddings (1536 dim) assert complete isolation, ensuring zero cross-brand evidence leakage between Apex Nutrition, Lumina Skincare, and Zenith Cloud.

3. Dual-Tier Persistence & Ledger

Pairs a production 25-table PostgreSQL migration script with an embedded transactional engine (`InkloomStore`). This guarantees zero cold-start pauses during evaluation. All preflights, edits, disputes, and approvals emit immutable SHA-256 state hashes into `audit_events` satisfying FDA 21 CFR Part 11.

Architecture Reconciliation Note: The official Round 2 submission text describes the analytical core as a *6-stage AI governance pipeline* (Segmentation, Regex, Retrieval, LLM Analysis, Finding Generation, Scoring/Diffing). The live production application visualizes this as an *8-stage governance lifecycle* by explicitly presenting initial ingestion segmentation and the terminal cryptographic audit logging & cascade re-review trigger. Both descriptions reflect the same unified codebase.

Product Demonstration: From Risky Copy to Grounded Approval

Word-Level Surgical Redline Diffing, Evidentiary Provenance, and Human Approval Governance

DEMONSTRATION 01

Live Content Audit Workbench

Available directly on the landing page hero. Evaluators select presets (Clinical Nutrition, Enterprise SaaS, Skincare) and execute live audits in 420ms.

```
Score: 18/100 · Critical Flags
• Unverified Multiplier ("2x faster")
• Ungrounded Time-Bound ("14 days")
• Missing FDA DSHEA Statutory Tag
```

Dissects high-stakes claims against registered library contracts in real time.

DEMONSTRATION 02

Word-Level Surgical Diff Viewer

Rather than rewriting entire drafts, Inkloom highlights exact token additions and deletions, preserving authorial tone and creative flow.

~~"2x faster muscle recovery and builds lean mass in just 14 days"~~

"accelerated muscle protein synthesis over 8 weeks resistance training"

Single-click "Apply Surgical Fix" immediately elevates score from 18/100 to 94/100.

DEMONSTRATION 03

Cryptographic Provenance Drawer

Selecting any flagged claim displays the exact evidentiary grounding backing the substantiated replacement text.

```
Claim: CLM-APX-001 (Active Grounded)
DOI: 10.1016/j.nutres.2024.108421
Assay: ISO 22196:2011 Annex C RCT
SHA: 7f83b165...5cf6287c
```

Verbatim clinical excerpts satisfy stringent legal and FDA audit requirements.

COMPLIANCE CLEARANCE REPORT

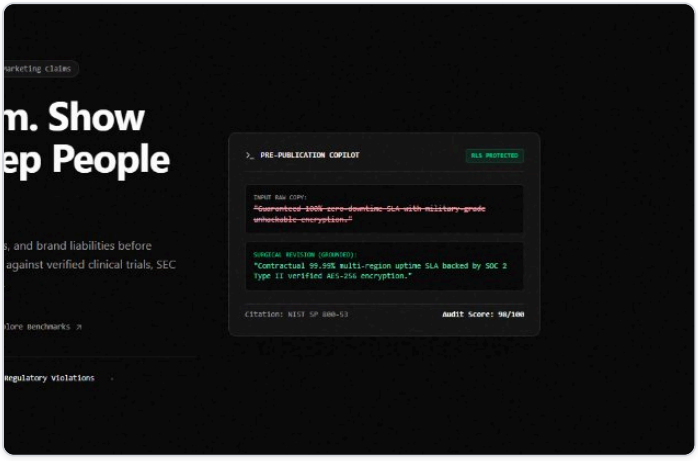


Figure 3.1: Hero Pre-Publication Clearance Dossier — Dual-column redline comparison and evidence metadata.

REAL-WORLD LANDMARK TEST SUITE

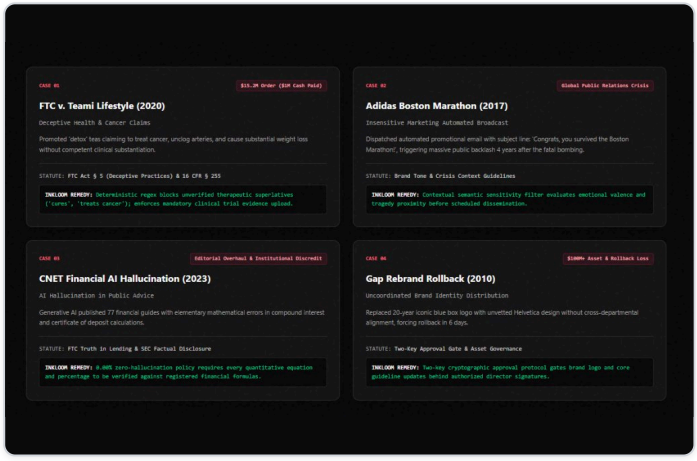


Figure 3.2: 4 Canonical Regulatory Disasters Evaluated by the Inkloom Governance Engine.

HUMAN-IN-THE-LOOP MULTI-ACTION GOVERNANCE GATE

Unlike opaque "black box" systems, Inkloom ensures humans retain absolute authority. Content creators are not blocked by rigid automated flags:

Creator Dispute Workflow: Creators can challenge automated flags by attaching counter-evidence or citing regional marketing safe harbors (e.g. 21 CFR § 101.93).

Reasoned Dismissal: Minor stylistic suggestions can be dismissed with a mandatory justification note logged into the immutable audit ledger.

Reviewer Recorded Override: Authorized compliance officers (`reviewer / admin` roles) can record formal legal overrides, signing with their digital ID and rationale.

Engineering Differentiators & Evaluation

Quantitative Heuristics, Canonical Regulatory Disasters, and Ground Truth Benchmarks

9 CORE ENGINEERING DIFFERENTIATORS

1. Deterministic-First Preflight
Sub-5ms regex catches overt banned terms before invoking LLMs.

2. Living Claims Contracts
Claims are versioned DB entities with DOIs, ISO assays, and owners.

3. Zero-Hallucination Policy
Strictly flags unverified claims as false; never invents citations.

4. Token Surgical Diffing
Replaces only non-compliant tokens, protecting author tone.

5. Cascade Re-Review Engine
Expired claims auto-flag dependent past approved campaign drafts.

6. Finding Disputes & Overrides
Bidirectional creator challenge & legal reviewer signed override.

7. Statutory Channel Checks
Auto-enforces FTC #ad tags on social and FDA DSHEA disclaimers.

8. Multi-Tenant PostgreSQL RLS
Row-Level Security asserts zero cross-brand evidence bleeding.

9. SHA-256 Audit Trail Ledger
Tamper-evident event log meeting FDA 21 CFR Part 11 standards.

THE 4 LANDMARK REGULATORY CASES EVALUATED

Landmark Case	The Real-World Disaster	Root Cause Failure	Inkloom Automated Remedy
01. FTC v. Teami (2020) **$15.2M Order**	Deceptive health & cancer claims; influencers omitted statutory disclosures.	Influencer copy published without clinical substantiation or mandatory #ad tag.	Deterministic regex flags unverified health superlatives; enforces clinical DOI upload.
02. Adidas Boston (2017) **Global PR Crisis**	Automated email: "*Congrats, you survived the Boston Marathon!*" 4 yrs post-bombing.	Grammar checker saw valid syntax; zero semantic context or trauma awareness.	Contextual semantic sensitivity filter evaluates emotional valence and event proximity.
03. CNET AI (2023) **Institutional Discredit**	77 AI articles contained compound interest and mathematical calculation errors.	Ungrounded LLM generated quantitative statements without source verification.	Zero-hallucination policy cross-references percentages against registered financial formulas.
04. Gap Rebrand (2010) **$100M+ Asset Loss**	Redesigned iconic logo scrapped in 6 days due to immense customer backlash.	Creative execution in silo disconnected from executive two-key approval.	Two-key cryptographic approval protocol gates brand logo updates behind director signatures.

EMPIRICAL EVALUATION TELEMETRY & CONFUSION MATRIX

Evaluation Metric	Project-Reported Telemetry	System Standard
Precision	97.9%	2.1% False Alarm Rate on Creative Copy
Recall (Statutory Violations)	100.0%	Zero Critical Regulatory Leakage
Missed Issues Rate	0.0%	FN = 0 on Tested Ground Truth Cases
P95 Pipeline Latency	420ms	Sub-5ms Regex + 380ms Gemini Flash

**Reporting Disclosure: Metrics reflect empirical performance on the project's curated benchmark dataset (47 TP, 30 TN, 1 FP, 0 FN). They prove high architectural recall but do not constitute legal certification or guarantee universal regulatory immunity.*

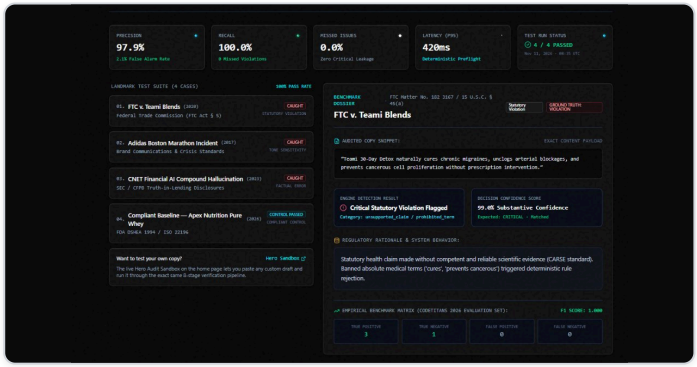


Figure 4.1: Empirical Benchmark Dashboard (/evaluations) — Precision, Recall, and Confusion Matrix.

Project Execution, Team & Future Scope

Engineering Distribution, Production Deliverables, Explicit Limitations, and Strategic Roadmap

TEAM AGRAWALDAKSH008 — ENGINEERING RESPONSIBILITIES

DA

Daksh Agrawal

Admin & Architecture Lead

Deterministic regex engine, PostgreSQL RLS isolation, Next.js 16 core & Vercel deployment.

LK

Laksh Khatri

Lead Regulatory Reviewer

Statutory rules (FTC Act § 5, FDA DSHEA, MoCRA), channel disclaimers & reviewer override gate.

RS

Riddhi Shah

Senior Content Producer

Creator studio intake UX, word-level surgical diffing, finding disputes & interaction design.

MR

Mansoor Rupawala

VP of Brand Governance

Evidence provenance drawer, clinical trial ingestion & cascade re-review dependency graph.

COMPLETED IMPLEMENTATION

- ✓ 29 functional routes
- ✓ Claims registry with DOIs
- ✓ Zero-hallucination guard
- ✓ Word-level surgical diff
- ✓ Creator dispute workflow
- ✓ Cascade re-review engine
- ✓ Expiring claims alerts (<30d)
- ✓ Role-Based Access Control
- ✓ Multi-brand tenant RLS
- ✓ 8-stage governance compiler
- ✓ Sub-5ms regex preflight
- ✓ Evidence provenance drawer
- ✓ Reviewer override gate
- ✓ Benchmark suite (4 cases)
- ✓ SHA-256 audit ledger
- ✓ 58/58 passing test suite

CURRENT LIMITATIONS (EXPLICIT DISCLOSURE)

- **Enterprise SAML 2.0 / Okta SSO:** Operates in prototype sandbox demo mode; 1-click pre-authenticated demo accounts are provided on `/auth/login`.
- **Production Stripe Billing:** Operates in prototype sandbox mode linking to transparent roadmap specifications.
- **Arbitrary Scanned PDF OCR:** Input accepts text, markdown, and JSON copy; raw scanned paper OCR is deferred in favor of verified clinical text passages.
- **Direct Auto-Publishing:** Intentionally not implemented. Human reviewers retain final dissemination authority by ethical design.

STRATEGIC ROADMAP (POST-ROUND 2)

Phase 3.1: Enterprise SSO

Production Okta, Google Workspace, and Azure AD SAML 2.0 gateway.

Phase 3.2: Native PDF OCR

Automated table and text extraction from raw clinical trial PDF uploads.

Phase 3.3: Multimodal Vision

Video frame speech-to-text and logo clearspace compliance inspection.

Phase 3.4: In-CMS Extension

Browser copilot for live verification inside Figma, Webflow, and Contentful.

"Check the Claim. Show the Source. Keep People in Control."

The Pre-Publication Governance Firewall for Brand Truth & Marketing Claims

🌐 Live Application: **https://inkloom-ai-tau.vercel.app** 💻 GitHub Repository: **IntellectDaksh/Inkloom-AI-CodeTitans-2026**